CSE 3023 Web Based Application Development

Lab Report 2

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Group: K1

Program:

Computer Science (Software Engineering)

DeleteServlet.java

```
package controller;


import model.Product;

import dao.ProductDao;

import java.io.IOException;

import java.io.PrintWriter;

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;


/**

 *

 * @author DING WEI QI
```

```
 */
public class DeleteServlet extends HttpServlet {

    private ProductDao productdao;

    public void init() {
        productdao = new ProductDao();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```
    protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet DeleteServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }
```

```
    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on
the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {


        int id = Integer.parseInt(request.getParameter("id"));
        productdao.deleteProduct(id); // Panggil DAO


        response.sendRedirect("ViewServlet");
```

```
}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}

/**
```

```
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>


}
```

InsertServlet.java

```
package controller;


import model.Product;

import dao.ProductDao;

import java.io.IOException;

import java.io.PrintWriter;

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;


/**

 *

 * @author DING WEI QI

 */

public class DeleteServlet extends HttpServlet {
```

```
private ProductDao productdao;



public void init() {

    productdao = new ProductDao();

}



/**

 * Processes requests for both HTTP <code>GET</code> and <code>POST</code>

 * methods.

 *

 * @param request servlet request

 * @param response servlet response

 * @throws ServletException if a servlet-specific error occurs

 * @throws IOException if an I/O error occurs

 */

protected void processRequest(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {
```

```
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet DeleteServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
```

```
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    productdao.deleteProduct(id); // Panggil DAO

    response.sendRedirect("ViewServlet");

}
```

```
/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}


/**
 * Returns a short description of the servlet.
 *
 * @return a String containing servlet description
```

```
     */

    @Override

    public String getServletInfo() {

        return "Short description";

    }// </editor-fold>



}
```

UpdateServlet.java

```
/*

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this license

 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this
template

 */

package controller;



import java.io.IOException;

import java.io.PrintWriter;

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

import model.Product;

import dao.ProductDao;
```

```
/**
 *
 * @author DING WEI QI
 */
public class UpdateServlet extends HttpServlet {


    private ProductDao productdao;


    public void init(){
        productdao = new ProductDao();
    }


    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
```

```
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
        /* TODO output your page here. You may use following sample code. */
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet UpdateServlet</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>Servlet UpdateServlet at " + request.getContextPath() + "</h1>");
        out.println("</body>");
        out.println("</html>");
    }
```

```
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on
the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
```

```
int id = Integer.parseInt(request.getParameter("id"));

Product existingProduct = productdao.selectProduct(id);

if (existingProduct != null) {
    out.println("<h2>Update Product (Using DAO)</h2>");
    out.println("<form action='UpdateServlet' method='POST'>");
    out.println("<input type='hidden' name='id' value='" + existingProduct.getId() + "'>");
    out.println("Name: <input type='text' name='name' value='" +
existingProduct.getName() + "' required><br><br>");
    out.println("Category: <input type='text' name='category' value='" +
existingProduct.getCategory() + "' required><br><br>");
    out.println("Price: <input type='text' name='price' value='" +
existingProduct.getPrice() + "' required><br><br>");
    out.println("Quantity: <input type='text' name='quantity' value='" +
existingProduct.getQuantity() + "' required><br><br>");
    out.println("<input type='submit' value='Update User'>");
    out.println("</form>");
```

```
    }
}


/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {


    int id = Integer.parseInt(request.getParameter("id"));


    String name = request.getParameter("name");
```

```
    String category = request.getParameter("category");

    String p = request.getParameter("price");

    String q = request.getParameter("quantity");



    double price = Double.parseDouble(p);

    int quantity = Integer.parseInt(q);



    Product user = new Product(id, name, category, price, quantity);

    productdao.updateProduct(user); // Panggil DAO



    response.sendRedirect("ViewServlet");

}



/**

 * Returns a short description of the servlet.

 *

 * @return a String containing servlet description

 */
```

```
    @Override

    public String getServletInfo() {

        return "Short description";

    }// </editor-fold>



}
```

ViewServlets.java

```
/*

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this license

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template

 */

package controller;



import java.io.IOException;

import java.io.PrintWriter;

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

import java.util.List;

import model.Product;

import dao.ProductDao;
```

```
/**
 *
 * @author DING WEI QI
 */
public class ViewServlet extends HttpServlet {


    private ProductDao productdao;


    public void init(){
        productdao = new ProductDao();
    }


    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
```

```
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
        /* TODO output your page here. You may use following sample code. */
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet ViewServlet</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>Servlet ViewServlet at " + request.getContextPath() + "</h1>");
        out.println("</body>");
        out.println("</html>");
```

```
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
```

```
PrintWriter out = response.getWriter();


List <Product> listProduct = productdao.selectAllProducts();


out.println("<h2>Product List (Using DAO)</h2>");
out.println("<table border='1'><tr><th>ID</th><th>Name</th>"
        + "<th>Category</th><th>Price</th><th>Quantity</th></tr>");


for (Product product : listProduct) {
    out.println("<tr>");
    out.println("<td>" + product.getId() + "</td>");
    out.println("<td>" + product.getName() + "</td>");
    out.println("<td>" + product.getCategory() + "</td>");
    out.println("<td>" + product.getPrice() + "</td>");
    out.println("<td>" + product.getQuantity() + "</td>");
    out.println("<td><a href='UpdateServlet?id=" + product.getId() + "'>Edit</a> | ");
    out.println("<a href='DeleteServlet?id=" + product.getId() + "'>Delete</a></td>");
    out.println("</tr>");
```

```
    }

    out.println("</table>");
    out.println("<br><a href='index.html'>Add New Product</a>");
}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
```

```
    }



    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>



}
```

ProductDao.java

```
/*
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this license
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 */
package dao;



/**
 *
 * @author DING WEI QI
 */
import model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
```

```
import java.util.ArrayList;

import java.util.List;



public class ProductDao {

    private String jdbcURL = "jdbc:mysql://localhost:3306/productinventorysystem";

    private String jdbcUsername = "root";

    private String jdbcPassword = "admin";

    private Connection connection;



    public ProductDao() {

        System.out.println("=== ProductDao Initialized ===");

        // Test connection immediately

        testConnectionOnStart();

    }



    private void testConnectionOnStart() {

        try (Connection conn = getConnection()) {

            if (conn != null && !conn.isClosed()) {
```

```
            System.out.println("✅ Database connection successful!");

            System.out.println("Connected to: " + conn.getMetaData().getURL());

            System.out.println("Database version: " +
conn.getMetaData().getDatabaseProductVersion());

        } else {

            System.out.println("❌ Failed to establish database connection");

        }

    } catch (SQLException e) {

        System.out.println("❌ Connection test failed: " + e.getMessage());

    }

}



protected Connection getConnection() {

    try {

        // Load driver (optional for newer JDBC versions)

        Class.forName("com.mysql.cj.jdbc.Driver");



        // Create connection
```

```
        Connection conn = DriverManager.getConnection(jdbcURL, jdbcUsername,
jdbcPassword);

        if (conn != null) {
            System.out.println("✅ Connection created successfully");
        }

        return conn;

    } catch (ClassNotFoundException e) {
        System.err.println("❌ MySQL Driver not found!");
        System.err.println("Add mysql-connector-java.jar to WEB-INF/lib");
        e.printStackTrace();
        return null;

    } catch (SQLException e) {
        System.err.println("❌ SQL Connection Error: " + e.getMessage());
```

```
        // Provide helpful error messages
        if (e.getMessage().contains("Access denied")) {
            System.err.println("→ Fix: Check your MySQL username and password");
            System.err.println("  Username: " + jdbcUsername);
            System.err.println("  Password: " + (jdbcPassword != null ? "****" : "null"));
        } else if (e.getMessage().contains("Unknown database")) {
            System.err.println("→ Fix: Database 'productinventorysystem' doesn't exist");
            System.err.println("  Run: CREATE DATABASE productinventorysystem;");
        } else if (e.getMessage().contains("Connection refused")) {
            System.err.println("→ Fix: MySQL server not reachable");
            System.err.println("  Check if MySQL is running on port 3306/3307");
            System.err.println("  Current URL: " + jdbcURL);
        }

        e.printStackTrace();
        return null;
    }
}
```

```
public void insertProduct(Product product){

    String sql = "INSERT INTO products (id, name, category, price, quantity) VALUES (?,
?, ?, ?, ?)";



    try (Connection conn = getConnection();

        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setInt(1, product.getId());

        pstmt.setString(2, product.getName());

        pstmt.setString(3, product.getCategory());

        pstmt.setDouble(4, product.getPrice());

        pstmt.setInt(5, product.getQuantity());

        pstmt.executeUpdate();

    } catch (SQLException e) {

        e.printStackTrace();

    }

}
```

```
    public Product selectProduct(int id){

        Product product = null;

        String sql = "SELECT id, name, category, price, quantity FROM products WHERE id
=?";



        try(Connection conn = getConnection();

            PreparedStatement pstmt = conn.prepareStatement(sql)){



            pstmt.setInt(1, id);

            ResultSet rs = pstmt.executeQuery();



            if(rs.next()){

                String name = rs.getString("name");

                String category = rs.getString("category");

                double price = rs.getDouble("price");

                int quantity = rs.getInt("quantity");

                product = new Product(id, name, category, price, quantity);

            }
```

```
    } catch (SQLException e) {
        e.printStackTrace();
    }

    return product;
}

public List<Product> selectAllProducts() {
    List<Product> products = new ArrayList<>();
    String sql = "SELECT * FROM products";

    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {
            ResultSet rs = pstmt.executeQuery();

            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
```

```
                String category = rs.getString("category");

                double price = rs.getDouble("price");

                int quantity = rs.getInt("quantity");

                products.add(new Product(id, name, category, price, quantity));

            }

        } catch (SQLException e) {

            e.printStackTrace();

        }



        return products;

    }



    public boolean updateProduct(Product product) {

        boolean rowUpdated = false;

        String sql = "UPDATE products SET name = ?, category = ?, price = ?, quantity = ?
WHERE id = ?";



        try (Connection conn = getConnection();
```

```
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

        pstmt.setString(1, product.getName());

        pstmt.setString(2, product.getCategory());

        pstmt.setDouble(3, product.getPrice());

        pstmt.setInt(4, product.getQuantity());

        pstmt.setInt(5, product.getId());



        rowUpdated = pstmt.executeUpdate() > 0;

    } catch (SQLException e) {

        e.printStackTrace();

    }

    return rowUpdated;

}



public boolean deleteProduct(int id) {

    boolean rowDeleted = false;

    String sql = "DELETE FROM products WHERE id = ?";
```

```
        try (Connection conn = getConnection();
            PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            rowDeleted = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}
```

Product.java

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change
this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package model;


/**
 *
 * @author DING WEI QI
 */


public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
```

```
private int quantity;



// Update & Delete

public Product (int id, String name, String category, double price, int quantity){

    this.id = id;

    this.name = name;

    this.category = category;

    this.price = price;

    this.quantity = quantity;

}



public Product(){}



// Insert

public Product(String name, String category, double price, int quantity){

    this.name = name;

    this.category = category;

    this.price = price;
```

```
    this.quantity = quantity;
}

public int getId() {
    return id;
}

public void setId(int id) {
    this.id = id;
}

public String getName() {
    return name;
}
```

Welcome to our system

Name: F790

Category: keyboard

Price: 199

Quantity: 23

Submit

```
public void setName(String name) {

    this.name = name;

}



public String getCategory() {

    return category;

}
```

```
public void setCategory(String category) {

    this.category = category;

}



public double getPrice() {

    return price;

}



public void setPrice(double price) {

    this.price = price;

}



public int getQuantity() {

    return quantity;

}



public void setQuantity(int quantity) {

    this.quantity = quantity;
```

```
    }

}
```

View product

Product List (Using DAO)

ID	Name	Category	Price	Quantity	
2	MSI B650	Motherboard	699.0	420	Edit | Delete
3	RTX 5060	GPU	2599.0	67	Edit | Delete
4	F790	keyboard	199.0	23	Edit | Delete

Add New Product

Update product

Update Product (Using DAO)

Name: F790

Category: keyboard

Price: 198.0

Quantity: 23

Update User

Product List (Using DAO)

ID	Name	Category	Price	Quantity	
2	MSI B650	Motherboard	699.0	420	Edit | Delete
3	RTX 5060	GPU	2599.0	67	Edit | Delete
4	F790	keyboard	198.0	23	Edit | Delete

Add New Product

Delete product

Product List (Using DAO)

ID	Name	Category	Price	Quantity	
2	MSI B650	Motherboard	699.0	420	Edit | Delete
4	F790	keyboard	198.0	23	Edit | Delete

Add New Product

Add new product

Product List (Using DAO)

ID	Name	Category	Price	Quantity	
2	MSI B650	Motherboard	699.0	420	Edit | Delete
4	F790	keyboard	198.0	23	Edit | Delete
5	9600x	cpu	1299.0	12	Edit | Delete

Add New Product